```
> alpha := proc(m, n)
    if n = 1 then
      (-1)^m;
    elif n = 2 then
      (-1)^(m + 1) * m^2;
    else
      (-1)^(m+n-1)·2^(n-2)·m·2^(3-2n)·(m+n-2)! / (pochhammer(1/2, n-1)·(m-n+1)!)
    end if;
  end proc:
```

$$\frac{(-1)^{m+n-1}\cdot 2^{n-2}\cdot m\cdot 2^{3-2n}\cdot (m+n-2)!}{\mathrm{pochhammer}\left(\frac{1}{2}, n-1\right)\cdot (m-n+1)!}$$

```
> V := unapply(sum(alpha(m, n) * (exp(I*y) / (I*y)^n + (-1)^(m + n) * exp(-I*y) / (I*y)^n), n = 1 .. m + 1), m, y);
```

$$V := (m, y) \mapsto -\frac{\mathrm{I}\cdot e^{\mathrm{I}\cdot y}\cdot (-1)^m\cdot \mathrm{hypergeom}\left([1, m, -m], \left[\frac{1}{2}\right], \frac{-\frac{\mathrm{I}}{2}}{y}\right)}{y} - \frac{\mathrm{I}\cdot e^{-\mathrm{I}\cdot y}\cdot (-1)^m\cdot (-1)^{m+1}\cdot \mathrm{hypergeom}\left([1, m, -m], \left[\frac{1}{2}\right], \frac{\frac{\mathrm{I}}{2}}{y}\right)}{y} \quad (1)$$

```
> tm := proc(m, y)
    piecewise(m = 0, 2 sin(y) / y, V(m, y));
  end proc:
```

$$piecewise\left(m = 0, \frac{2\sin(y)}{y}, V(m, y)\right);$$

```
>
> X := (n, y)→simplify(int(ChebyshevT(n, x)·exp(-I·x·y), x = -1 ..1)) :
> Ynorm := unapply(sqrt(4 (-1)^m Pi (2 m^2 - 1) / (4 m^2 - 1)), m);
```

$$Ynorm := m \mapsto 2\cdot\sqrt{\frac{(-1)^m\cdot \pi\cdot (2\cdot m^2 - 1)}{4\cdot m^2 - 1}} \quad (2)$$

```
> Y := unapply(simplify(tm(m, y) / Ynorm(m)), m, y) :
    # Normalized Fourier transformed Type-I Chebyshev Polynomials
> Matrix(map(simplify, [seq([(X(m, y)) - (tm(m, y))], m = 0 ..4)]))
```

(3)

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{3}$$

```
> evalf([seq(Ynorm(i, y), i = 1 ..20)])
```

$$[2.046653416\,I, 2.421632979, 2.470559800\,I, 2.486654823, 2.493936404\,I, 2.497848464, 2.500192762\,I, 2.501708489, 2.502745038\,I, 2.503485167, 2.504032078\,I, 2.504447651, 2.504770824\,I, 2.505027106, 2.505233766\,I, 2.505402839, 2.505542920\,I, 2.505660277, 2.505759576\,I, 2.505844344] \tag{4}$$

```
>
> f := unapply(int(BesselJ(0, y)·exp(−I·x·y), y = 0 ..infinity) , x) assuming x ≠ 0
```

$$f := x \mapsto \frac{1}{\sqrt{-x^2+1}} \tag{5}$$

```
> # Gram-Schmidt process for the Fourier transforms of Chebyshev polynomials
  GramSchmidt := proc(vecList)
  local n, m, inner, projection, u, v, orthoList;
  orthoList := [ ];
  for n from 1 to numelems(vecList) do
     v := vecList[n];
     projection := 0;
     for m from 1 to n − 1 do
        u := orthoList[m];
        inner := int(u * v, y = −infinity .. infinity);
        projection := projection + inner * u;
     end do;
     u := simplify(v − projection);
     u := simplify(u / sqrt(int(u * conjugate(u), y = −infinity .. infinity)));
     orthoList := [op(orthoList), u];
  end do;
  return orthoList;
   end proc:
> # Generate the functions
  funcList := map(simplify, [seq(Y(n, y), n = 0 .. 12)]) :

> # Apply Gram-Schmidt process
  orthoFuncs := GramSchmidt(funcList) :

> ?Legendre
>
```

> $\left(\left[seq\left(lcoeff\left(simplify\left(\frac{\exp\left(\frac{I\cdot \mathrm{Pi}\cdot(2\,n+1)}{2}\right)\cdot \mathrm{sqrt}(\mathrm{Pi})}{\mathrm{sqrt}(2\,n-1)}\,inttrans[invfourier](\mathrm{psi}[n], y, t)\right)\right), n = 1..13\right)\right]\right)$

$$\left[\frac{\mathrm{I}}{2}, -\frac{1}{2}, \frac{3\,\mathrm{I}}{4}, -\frac{5}{4}, \frac{35\,\mathrm{I}}{16}, -\frac{63}{16}, \frac{231\,\mathrm{I}}{32}, -\frac{429}{32}, \frac{6435\,\mathrm{I}}{256}, -\frac{12155}{256}, \frac{46189\,\mathrm{I}}{512}, -\frac{88179}{512}, \frac{676039\,\mathrm{I}}{2048}\right] \quad \textbf{(6)}$$

> $LRETools{:}\text{-}GuessRecurrence(\mathbf{(6)}, u(n))$

> $gfun[guessgf]\,(\mathbf{(6)}, x)$

> $gfun{:}\text{-}guesseqn(\mathbf{(6)}, g(x))$

$$\left[\left\{\left(4\,x^2-1\right)\left(\frac{d^2}{dx^2}\,g(x)\right)+12\,x\left(\frac{d}{dx}\,g(x)\right)+3\,g(x), g(0)=\frac{\mathrm{I}}{2}, \mathrm{D}(g)(0)=-\frac{1}{2}\right\}, ogf\right] \quad \textbf{(7)}$$

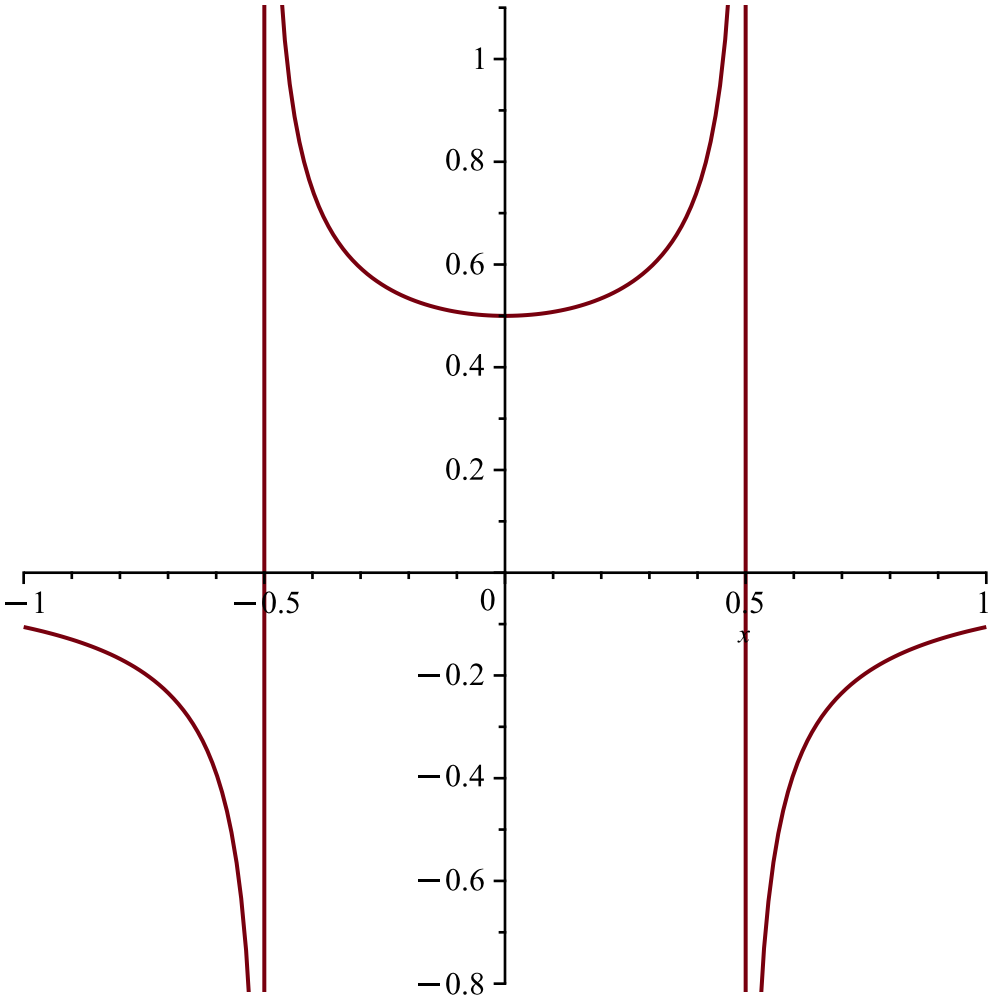
> $gen := unapply((op(2, dsolve(op(1, \mathbf{(7)})))), x)$

$$gen := x \mapsto \frac{\frac{1}{4}+\frac{\mathrm{I}}{4}}{\sqrt{2\cdot x+1}} - \frac{\frac{1}{4}+\frac{\mathrm{I}}{4}}{\sqrt{2\cdot x-1}} \quad \textbf{(8)}$$

> $series(gen(x), x)$ assuming $x > 0$

$$\frac{\mathrm{I}}{2} - \frac{1}{2}\,x + \frac{3\,\mathrm{I}}{4}\,x^2 - \frac{5}{4}\,x^3 + \frac{35\,\mathrm{I}}{16}\,x^4 - \frac{63}{16}\,x^5 + \mathrm{O}(x^6) \quad \textbf{(9)}$$

> $plot([\mathrm{Im}(gen(x))], x = -1..1)$



> $thing := unapply\left(simplify\left(eval\left(\frac{diff(gen(x), x\$n)}{n!}, x=0\right)\right), n\right);$

$$thing := n \mapsto \frac{\left(\frac{1}{4}+\frac{\mathrm{I}}{4}\right)\cdot \mathrm{pochhammer}\left(\frac{1}{2}-n, n\right)\cdot\left(\mathrm{I}\cdot(-1)^{-n}+1\right)\cdot 2^{n}}{n!} \tag{10}$$

> $seq(thing(n), n=1..5)$

$$-\frac{1}{2}, \frac{3\,\mathrm{I}}{4}, -\frac{5}{4}, \frac{35\,\mathrm{I}}{16}, -\frac{63}{16} \tag{11}$$

> $simplify\left(sum\left(thing(n)\cdot\frac{x^n}{n!}, n=0..\mathrm{infinity}\right)\right)$

$$\frac{\mathrm{BesselI}(0, x)\left((1+\mathrm{I})\,\mathrm{e}^{-x}+(-1+\mathrm{I})\,\mathrm{e}^{x}\right)}{4} \tag{12}$$

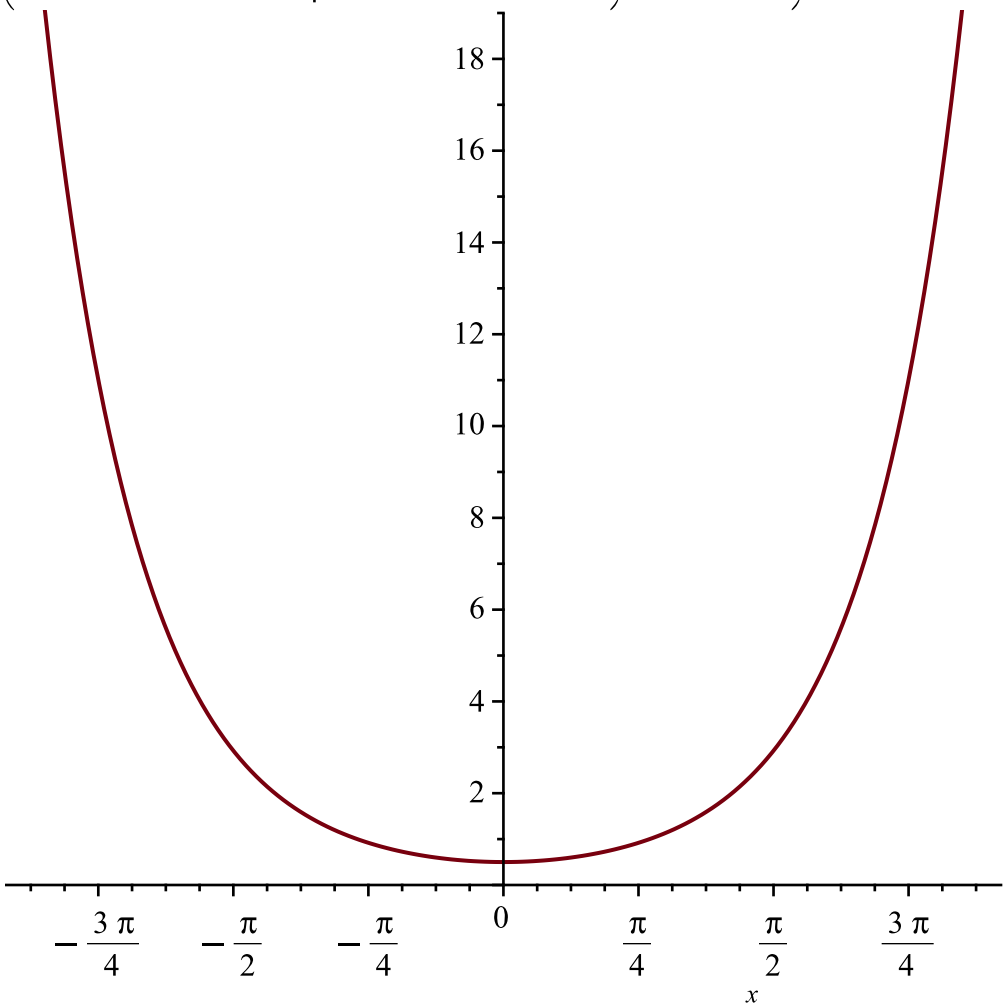
> $series(\mathbf{(12)}, x)$

$$\frac{\mathrm{I}}{2}-\frac{1}{2}x+\frac{3\,\mathrm{I}}{8}x^2-\frac{5}{24}x^3+\frac{35\,\mathrm{I}}{384}x^4-\frac{21}{640}x^5+\mathrm{O}(x^6) \tag{13}$$

> $limit(diff(\mathbf{(12)}, x\$2), x=0)$

$$\frac{3\,\mathrm{I}}{4} \tag{14}$$

> $plot\left(\text{abs}\left(\frac{\text{BesselI}(0, x)\ \left((1+\text{I})\ e^{-x} + (-1+\text{I})\ e^{x}\right)}{4}\right), x = -\text{Pi}\,..\text{Pi}\right)$



> ?*guesseqn*

> ?*guess*

> *series*((1 − *x*)^(−3/2), *x*, 10)

$$1 + \frac{3}{2}x + \frac{15}{8}x^2 + \frac{35}{16}x^3 + \frac{315}{128}x^4 + \frac{693}{256}x^5 + \frac{3003}{1024}x^6 + \frac{6435}{2048}x^7 + \frac{109395}{32768}x^8 + \frac{230945}{65536}x^9 + O(x^{10}) \quad \textbf{(15)}$$

> *seq*(binomial(2 *n*, *n*) / 4^*n*, *n* = 1 ..13)

$$\frac{1}{2}, \frac{3}{8}, \frac{5}{16}, \frac{35}{128}, \frac{63}{256}, \frac{231}{1024}, \frac{429}{2048}, \frac{6435}{32768}, \frac{12155}{65536}, \frac{46189}{262144}, \frac{88179}{524288}, \frac{676039}{4194304}, \frac{1300075}{8388608} \quad \textbf{(16)}$$

> *e* := **proc**(*l*, *m*) **local** *k*; *sum*(2^(*k*−2 * *m*) * binomial(2 * *m*−2 * *k*, *m*−*k*) * binomial(*m* + *k*, *m*) * binomial(*k*, *l*), *k* = *l*..*m*); **end**;

e := **proc**(*l*, *m*) **(17)**

local *k*;

$sum(2\text{^}(k-2*m)*\text{binomial}(2*m-2*k, m-k)*\text{binomial}(m+k, m)*\text{binomial}(k, l), k=l..m)$

end proc

> $e(m, m)$

$$2^{-m}\binom{2m}{m} \tag{18}$$

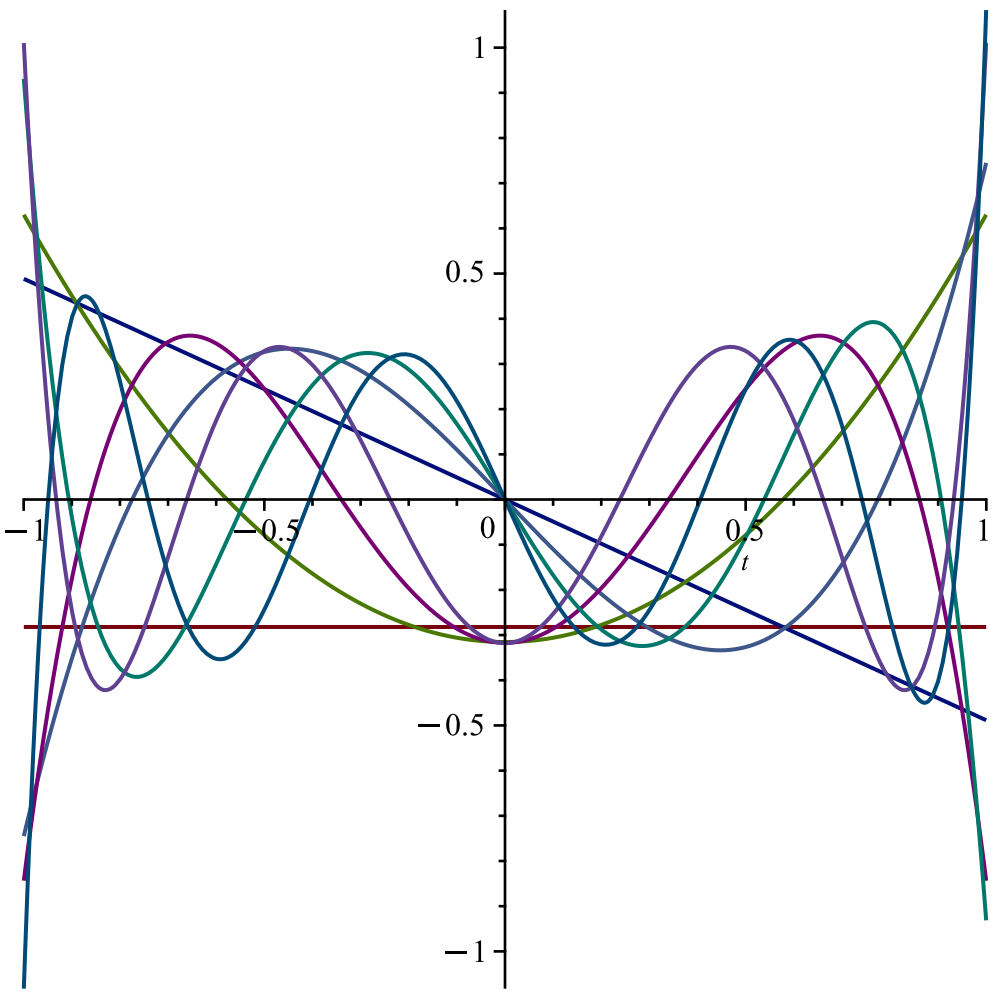
> $eval\left(2^{-m}\binom{2m}{m}, m=m+1\right)$

$$2^{-1-m}\binom{2m+2}{m+1} \tag{19}$$

> $seq(\mathbf{(19)}, m=1..10)$

$$\frac{3}{2}, \frac{5}{2}, \frac{35}{8}, \frac{63}{8}, \frac{231}{16}, \frac{429}{16}, \frac{6435}{128}, \frac{12155}{128}, \frac{46189}{256}, \frac{88179}{256} \tag{20}$$

> $plot(\mathbf{(6)}, t=-1..1)$



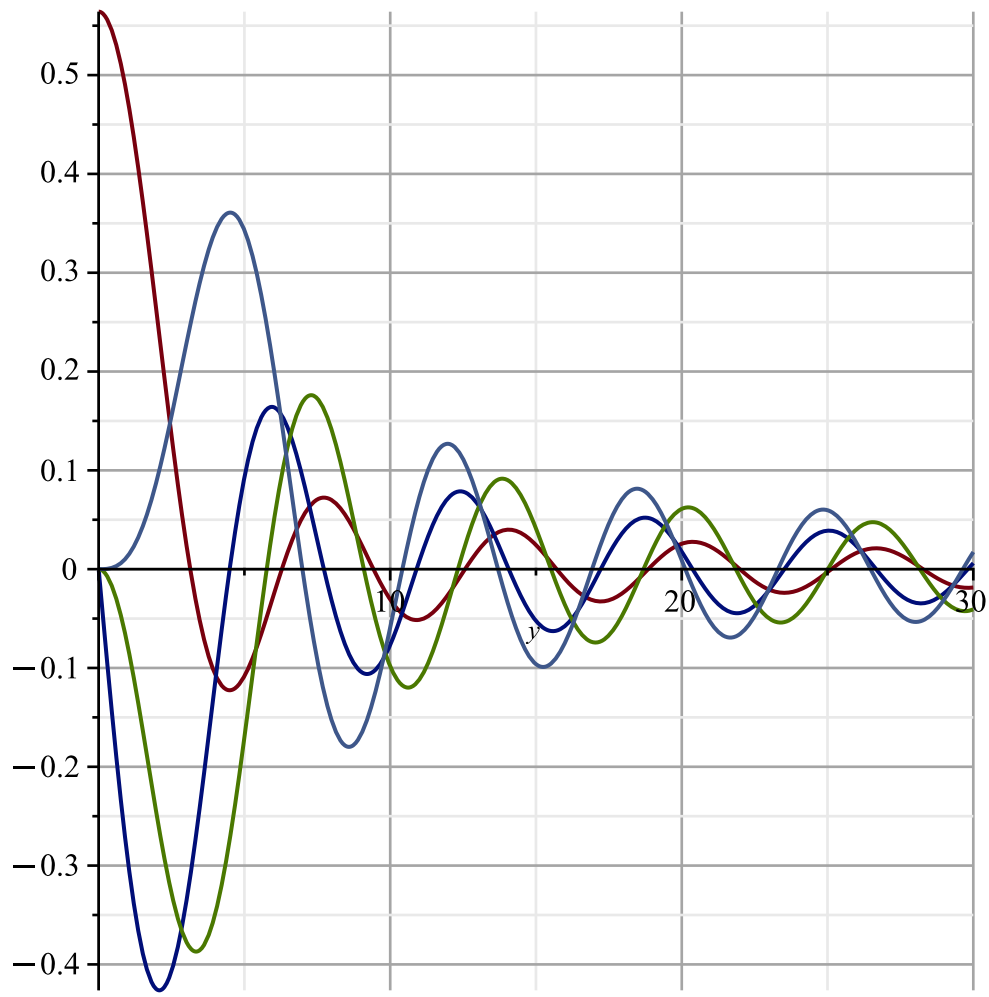
> $Vector([seq(simplify(inttrans[invfourier](\text{psi}[n], y, t)), n=1..8)])$

$$\left[\begin{array}{c}
\frac{-\mathrm{Heaviside}(t-1)+\mathrm{Heaviside}(t+1)}{2\sqrt{\pi}} \\
\frac{\frac{\mathrm{I}}{2}\sqrt{3}\,t\,(\mathrm{Heaviside}(t-1)-\mathrm{Heaviside}(t+1))}{\sqrt{\pi}} \\
-\frac{3\,(\mathrm{Heaviside}(t-1)-\mathrm{Heaviside}(t+1))\left(t^2-\frac{1}{3}\right)\sqrt{5}}{4\sqrt{\pi}} \\
\frac{\frac{5\,\mathrm{I}}{4}\,(\mathrm{Heaviside}(t-1)-\mathrm{Heaviside}(t+1))\left(t^2-\frac{3}{5}\right)\sqrt{7}\,t}{\sqrt{\pi}} \\
-\frac{3\,(35\,t^4-30\,t^2+3)\,(\mathrm{Heaviside}(t-1)-\mathrm{Heaviside}(t+1))}{16\sqrt{\pi}} \\
\frac{\frac{\mathrm{I}}{16}\sqrt{11}\,t\,(63\,t^4-70\,t^2+15)\,(\mathrm{Heaviside}(t-1)-\mathrm{Heaviside}(t+1))}{\sqrt{\pi}} \\
-\frac{\sqrt{13}\,(231\,t^6-315\,t^4+105\,t^2-5)\,(\mathrm{Heaviside}(t-1)-\mathrm{Heaviside}(t+1))}{32\sqrt{\pi}} \\
\frac{\frac{\mathrm{I}}{32}\sqrt{15}\,t\,(429\,t^6-693\,t^4+315\,t^2-35)\,(\mathrm{Heaviside}(t-1)-\mathrm{Heaviside}(t+1))}{\sqrt{\pi}}
\end{array}\right] \tag{21}$$

> psi[1..4]

$$\left[\begin{array}{c}
\frac{\sin(y)}{\sqrt{\pi}\,y} \\
-\frac{\sqrt{3}\,\sin(y)}{\sqrt{\pi}\,y^2}+\frac{\sqrt{3}\,\cos(y)}{\sqrt{\pi}\,y} \\
\frac{\sqrt{5}\,(y^2-3)\,\sin(y)}{y^3\sqrt{\pi}}+\frac{3\sqrt{5}\,\cos(y)}{y^2\sqrt{\pi}} \\
\frac{\sqrt{7}\,(-6\,y^2+15)\,\sin(y)}{\sqrt{\pi}\,y^4}+\frac{\sqrt{7}\,(y^3-15\,y)\,\cos(y)}{\sqrt{\pi}\,y^4}
\end{array}\right] \tag{22}$$

> *plot*(psi[1..4], *y* = 0..30, *gridlines* = *true*)




```
> # Print orthogonalized functions
  psi := map(z→ collect(z, sin(y)), Vector(orthoFuncs)) :
> psi[1..9]
```

$$\left[\left[\frac{\sin(y)}{\sqrt{\pi}\, y}\right], \left[-\frac{\sqrt{3}\,\sin(y)}{\sqrt{\pi}\, y^2} + \frac{\sqrt{3}\,\cos(y)}{\sqrt{\pi}\, y}\right], \left[\frac{\sqrt{5}\,(y^2-3)\,\sin(y)}{y^3\sqrt{\pi}} + \frac{3\sqrt{5}\,\cos(y)}{y^2\sqrt{\pi}}\right], \left[\frac{\sqrt{7}\,(-6\,y^2+15)\,\sin(y)}{\sqrt{\pi}\, y^4} + \frac{\sqrt{7}\,(y^3-15\,y)\,\cos(y)}{\sqrt{\pi}\, y^4}\right], \right. \tag{23}$$

$$\left[\frac{3\left(y^4-45y^2+105\right)\sin(y)}{\sqrt{\pi}\,y^5}+\frac{3\left(10y^3-105y\right)\cos(y)}{\sqrt{\pi}\,y^5}\right],$$

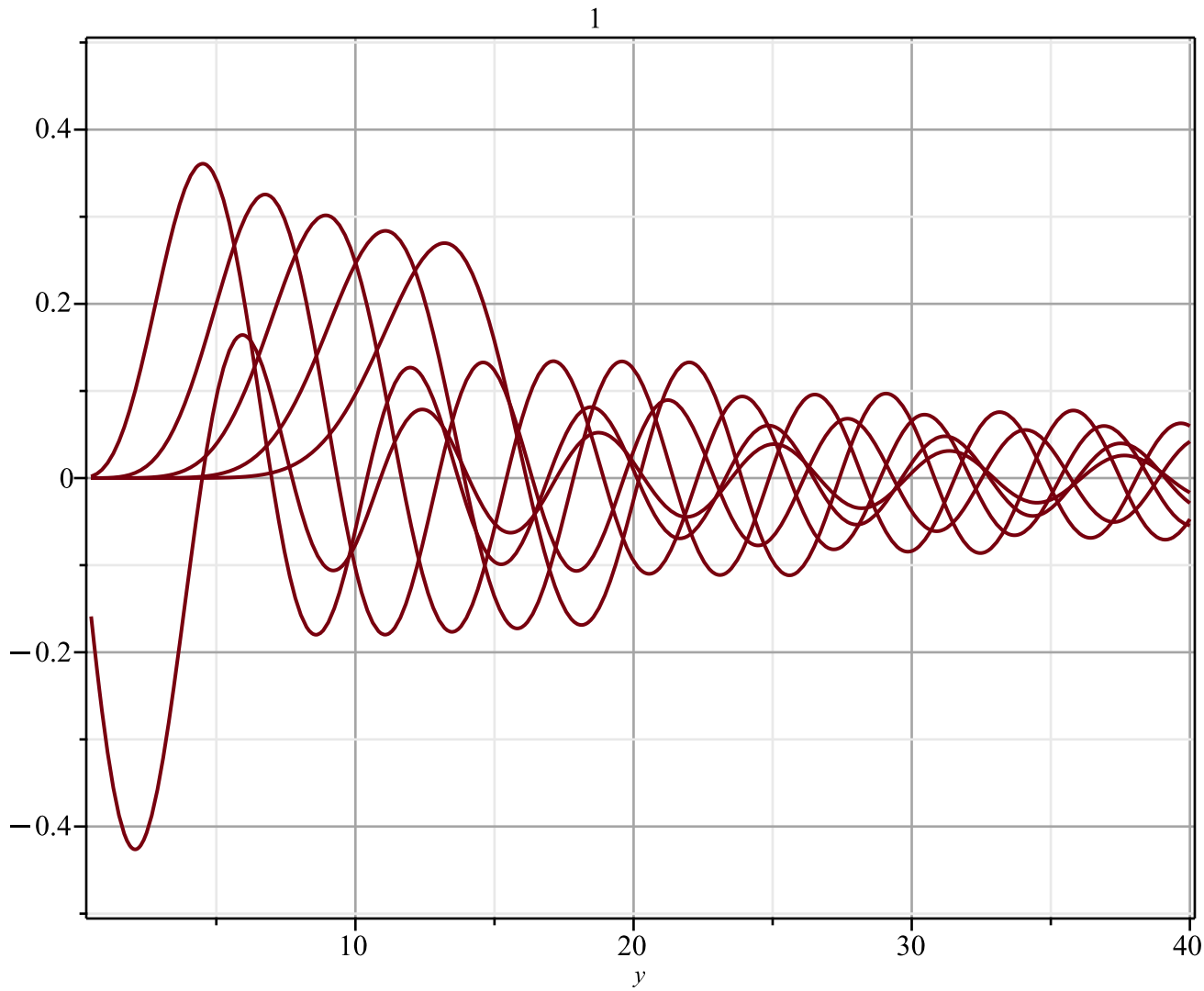
$$\left[\frac{\sqrt{11}\left(-15y^4+420y^2-945\right)\sin(y)}{\sqrt{\pi}\,y^6}+\frac{\sqrt{11}\left(y^5-105y^3+945y\right)\cos(y)}{\sqrt{\pi}\,y^6}\right],$$

$$\left[\frac{\sqrt{13}\left(y^6-210y^4+4725y^2-10395\right)\sin(y)}{\sqrt{\pi}\,y^7}+\frac{21\sqrt{13}\cos(y)\left(y^4-60y^2+495\right)}{\sqrt{\pi}\,y^6}\right],$$

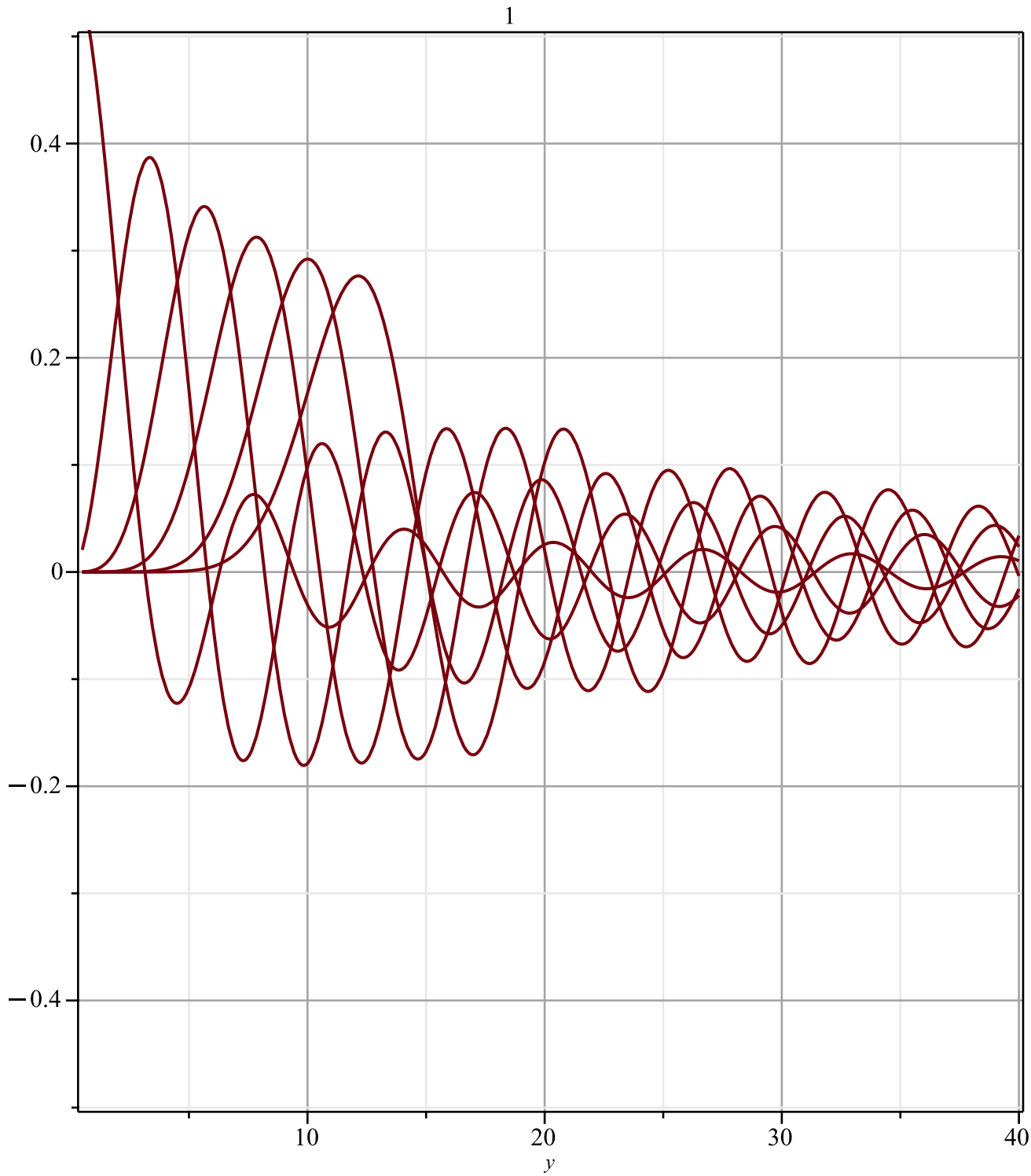
$$\left[\frac{\sqrt{15}\left(-28y^6+3150y^4-62370y^2+135135\right)\sin(y)}{\sqrt{\pi}\,y^8}+\frac{\sqrt{15}\left(y^7-378y^5+17325y^3-135135y\right)\cos(y)}{\sqrt{\pi}\,y^8}\right],$$

$$\left[\frac{\sqrt{17}\left(y^8-630y^6+51975y^4-945945y^2+2027025\right)\sin(y)}{\sqrt{\pi}\,y^9}+\frac{36\sqrt{17}\left(y^6-\frac{385}{2}y^4+\frac{15015}{2}y^2-\frac{225225}{4}\right)\cos(y)}{\sqrt{\pi}\,y^8}\right]\Bigg]$$

```
>
> plots[display](seq(plot(((-1)^n · psi[2 n]), y = 0.5 ..40, title = n, view = [0.5 ..40, -.5 ..0.5],
    gridlines, axes = boxed), n = 1 ..6))
```



> $plots[display](seq(plot(((-1)^{n+1} \cdot \mathrm{psi}[2\,n-1]), y = 0.5\,..40, title = n, view = [0.5\,..40, -.5\,..0.5], gridlines, axes = boxed), n = 1\,..6))$



>

>

> $hmmsin := map\left(z \rightarrow \left(\frac{op(1, z) \cdot \mathrm{sqrt}(\mathrm{Pi})}{\sin(y)}\right), \mathrm{psi}\right) : coolsin := Vector\left(\left[seq\left(simplify\left(\frac{hmmsin[i] \cdot y^{i}}{\mathrm{sqrt}(2 \cdot i - 1)}\right), i = 1 .. 13\right)\right]\right)$

$coolsin := \left[\left[\sqrt{\pi}\, y\right],\right.$ **(24)**

$$[1],$$
$$[y^2 - 3],$$
$$[-6\,y^2 + 15],$$
$$[y^4 - 45\,y^2 + 105],$$
$$[-15\,y^4 + 420\,y^2 - 945],$$
$$[y^6 - 210\,y^4 + 4725\,y^2 - 10395],$$
$$[-28\,y^6 + 3150\,y^4 - 62370\,y^2 + 135135],$$
$$[y^8 - 630\,y^6 + 51975\,y^4 - 945945\,y^2 + 2027025],$$
$$[-45\,y^8 + 13860\,y^6 - 945945\,y^4 + 16216200\,y^2 - 34459425],$$
$$[y^{10} - 1485\,y^8 + 315315\,y^6 - 18918900\,y^4 + 310134825\,y^2 - 654729075],$$
$$[-66\,y^{10} + 45045\,y^8 - 7567560\,y^6 + 413513100\,y^4 - 6547290750\,y^2 + 13749310575],$$
$$[y^{12} - 3003\,y^{10} + 1351350\,y^8 - 192972780\,y^6 + 9820936125\,y^4 - 151242416325\,y^2 + 316234143225]]$$

> $hmmcos := map\left(z \rightarrow \left(\frac{op(2, z) \cdot \mathrm{sqrt}(\mathrm{Pi})}{\cos(y)}\right), \mathrm{psi}\right) : coolcos := Vector\left(\left[seq\left(simplify\left(\frac{hmmcos[i] \cdot y^{i-1}}{\mathrm{sqrt}(2 \cdot i - 1)}\right), i = 1..13\right)\right]\right)$

$coolcos :=$ **(25)**

$$\left[\begin{array}{c}
\frac{\sqrt{\pi}\,\sec(y)}{y} \\
-1 \\
3 \\
y^2-15 \\
10\,y^2-105 \\
y^4-105\,y^2+945 \\
21\,y^4-1260\,y^2+10395 \\
y^6-378\,y^4+17325\,y^2-135135 \\
36\,y^6-6930\,y^4+270270\,y^2-2027025 \\
y^8-990\,y^6+135135\,y^4-4729725\,y^2+34459425 \\
55\,y^8-25740\,y^6+2837835\,y^4-91891800\,y^2+654729075 \\
y^{10}-2145\,y^8+675675\,y^6-64324260\,y^4+1964187225\,y^2-13749310575 \\
78\,y^{10}-75075\,y^8+18378360\,y^6-1571349780\,y^4+45831035250\,y^2-316234143225
\end{array}\right]$$

> $cosMatrix := convert(convert(map(z \rightarrow ListTools{:-}Reverse(convert(MTM{:-}coeffs(z, y), list)), coolcos[3..12]), list), Matrix)$

$$cosMatrix := \left[\begin{array}{cccccc}
3 & 0 & 0 & 0 & 0 & 0 \\
-15 & 1 & 0 & 0 & 0 & 0 \\
-105 & 10 & 0 & 0 & 0 & 0 \\
945 & -105 & 1 & 0 & 0 & 0 \\
10395 & -1260 & 21 & 0 & 0 & 0 \\
-135135 & 17325 & -378 & 1 & 0 & 0 \\
-2027025 & 270270 & -6930 & 36 & 0 & 0 \\
34459425 & -4729725 & 135135 & -990 & 1 & 0 \\
654729075 & -91891800 & 2837835 & -25740 & 55 & 0 \\
-13749310575 & 1964187225 & -64324260 & 675675 & -2145 & 1
\end{array}\right] \tag{26}$$

> $Vector([seq(simplify(\mathrm{LegendreP}(n, t) \cdot n!), n = 1..9)])$

(27)

$$\begin{bmatrix} t \\ 3\,t^2 - 1 \\ 15\,t^3 - 9\,t \\ 105\,t^4 - 90\,t^2 + 9 \\ 945\,t^5 - 1050\,t^3 + 225\,t \\ 10395\,t^6 - 14175\,t^4 + 4725\,t^2 - 225 \\ 135135\,t^7 - 218295\,t^5 + 99225\,t^3 - 11025\,t \\ 2027025\,t^8 - 3783780\,t^6 + 2182950\,t^4 - 396900\,t^2 + 11025 \\ 34459425\,t^9 - 72972900\,t^7 + 51081030\,t^5 - 13097700\,t^3 + 893025\,t \end{bmatrix} \tag{27}$$

```
>
> sinMatrix := convert(convert(map(z→ListTools:-Reverse(convert(MTM:-coeffs(z, y), list)),
    coolsin[3 ..12]), list), Matrix)
```

$$sinMatrix := \begin{bmatrix} -3 & 1 & 0 & 0 & 0 & 0 \\ 15 & -6 & 0 & 0 & 0 & 0 \\ 105 & -45 & 1 & 0 & 0 & 0 \\ -945 & 420 & -15 & 0 & 0 & 0 \\ -10395 & 4725 & -210 & 1 & 0 & 0 \\ 135135 & -62370 & 3150 & -28 & 0 & 0 \\ 2027025 & -945945 & 51975 & -630 & 1 & 0 \\ -34459425 & 16216200 & -945945 & 13860 & -45 & 0 \\ -654729075 & 310134825 & -18918900 & 315315 & -1485 & 1 \\ 13749310575 & -6547290750 & 413513100 & -7567560 & 45045 & -66 \end{bmatrix} \tag{28}$$

```
> B := (n, k) → pochhammer(n + 1, k) * binomial(n, k) / 2^k
```

$$B := (n, k) \mapsto \frac{\text{pochhammer}(n+1, k) \cdot \binom{n}{k}}{2^k} \tag{29}$$

```
> C := unapply(simplify( (−1)^(n+1) · (−1)^binomial(k, 2) · B(k + 1, k − (2 n − 1))), n, k)
    assuming n :: nonnegint, k :: nonnegint
```

$$C := (n, k) \mapsto -(-1)^{n + \binom{k}{2}} \cdot 2^{-1-k+2\cdot n} \cdot \binom{k+1}{k-2\cdot n+1} \cdot \text{pochhammer}(k+2, k-2\cdot n+1) \tag{30}$$

```
> BP := unapply(sum((n + k)!* (x/2)^k/((n − k)!* k!), k = 0 .. n), n, x)
```

$$BP := (n, x) \mapsto \text{hypergeom}\left([-n, n+1], [\ ], -\frac{x}{2}\right) \tag{31}$$

```
>
> Matrix([seq([coeffs(simplify(BP(n, x)), x)], n = 2 ..8)])
```

$$\begin{bmatrix} 3 & 3 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 15 & 15 & 6 & 1 & 0 & 0 & 0 & 0 & 0 \\ 105 & 105 & 45 & 10 & 1 & 0 & 0 & 0 & 0 \\ 945 & 945 & 420 & 105 & 15 & 1 & 0 & 0 & 0 \\ 10395 & 10395 & 4725 & 1260 & 210 & 21 & 1 & 0 & 0 \\ 135135 & 135135 & 62370 & 17325 & 3150 & 378 & 28 & 1 & 0 \\ 2027025 & 2027025 & 945945 & 270270 & 51975 & 6930 & 630 & 36 & 1 \end{bmatrix} \quad (32)$$

>

> $convert([seq(flist(C(n,k), n=0..k), k=1..10)], Matrix);$
  # *first col is constants, 2nd col is coeffs of x, 3rd is of* x^2 ...

$$\begin{bmatrix} -3 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 15 & -6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 105 & -45 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -945 & 420 & -15 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -10395 & 4725 & -210 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 135135 & -62370 & 3150 & -28 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2027025 & -945945 & 51975 & -630 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -34459425 & 16216200 & -945945 & 13860 & -45 & 0 & 0 & 0 & 0 & 0 & 0 \\ -654729075 & 310134825 & -18918900 & 315315 & -1485 & 1 & 0 & 0 & 0 & 0 & 0 \\ 13749310575 & -6547290750 & 413513100 & -7567560 & 45045 & -66 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (33)$$

> $simplify(BP(6,x))$

$$10395\,x^6 + 10395\,x^5 + 4725\,x^4 + 1260\,x^3 + 210\,x^2 + 21\,x + 1 \quad (34)$$

> $woo := (m,n) \rightarrow (-1)^{2n-m} \cdot (-1)$^binomial$(n-1,2) \cdot (2*n-m)!/(m!*(n-m)!*2^{(n-m)})$; # *This is a column generator, unlike C which is a row-generator*

$$woo := (m,n) \mapsto \frac{(-1)^{2\cdot n-m} \cdot (-1)^{\binom{n-1}{2}} \cdot (2\cdot n-m)!}{m!\cdot(n-m)!\cdot 2^{n-m}} \quad (35)$$

> $wow := unapply(woo(m+1, n+2), m, n);$

$$wow := (m,n) \mapsto \frac{(-1)^{3+2\cdot n-m} \cdot (-1)^{\binom{n+1}{2}} \cdot (3+2\cdot n-m)!}{(m+1)!\cdot(1+n-m)!\cdot 2^{1+n-m}} \quad (36)$$

> $convert([seq(flist(woo(2\cdot(m+1)+1, n), m=1..4), n=1..4)], Matrix);$
  # *first col is constants, 2nd col is coeffs of x, 3rd is of* x^2 ...

Error, (in flist) numeric exception: division by zero

> $seq(limit(bah(0,n), n=k), k=0..10)$

$$1, 1, 3, 15, 105, 945, 10395, 135135, 2027025, 34459425, 654729075 \quad (37)$$

> $seq(limit(bah(2,n), n=k), k=0..10)$

$$1, 0, 1, 6, 45, 420, 4725, 62370, 945945, 16216200, 310134825 \quad (38)$$

> $seq(limit(bah(4, n), n = k), k = 0..10)$

$$\frac{1}{3}, -\frac{1}{3}, 0, 0, 1, 15, 210, 3150, 51975, 945945, 18918900 \quad (39)$$

> $seq(limit(bah(6, n), n = k), k = 0..10)$

$$\frac{2}{45}, -\frac{4}{45}, \frac{1}{15}, 0, 0, 0, 1, 28, 630, 13860, 315315 \quad (40)$$

> $sinMatrix$

$$\begin{bmatrix} -3 & 1 & 0 & 0 & 0 & 0 \\ 15 & -6 & 0 & 0 & 0 & 0 \\ 105 & -45 & 1 & 0 & 0 & 0 \\ -945 & 420 & -15 & 0 & 0 & 0 \\ -10395 & 4725 & -210 & 1 & 0 & 0 \\ 135135 & -62370 & 3150 & -28 & 0 & 0 \\ 2027025 & -945945 & 51975 & -630 & 1 & 0 \\ -34459425 & 16216200 & -945945 & 13860 & -45 & 0 \\ -654729075 & 310134825 & -18918900 & 315315 & -1485 & 1 \\ 13749310575 & -6547290750 & 413513100 & -7567560 & 45045 & -66 \end{bmatrix} \quad (41)$$

>

> *# Now to find the function for the cosMatrix*

> $cosMatrix$

$$\begin{bmatrix} 3 & 0 & 0 & 0 & 0 & 0 \\ -15 & 1 & 0 & 0 & 0 & 0 \\ -105 & 10 & 0 & 0 & 0 & 0 \\ 945 & -105 & 1 & 0 & 0 & 0 \\ 10395 & -1260 & 21 & 0 & 0 & 0 \\ -135135 & 17325 & -378 & 1 & 0 & 0 \\ -2027025 & 270270 & -6930 & 36 & 0 & 0 \\ 34459425 & -4729725 & 135135 & -990 & 1 & 0 \\ 654729075 & -91891800 & 2837835 & -25740 & 55 & 0 \\ -13749310575 & 1964187225 & -64324260 & 675675 & -2145 & 1 \end{bmatrix} \quad (42)$$

> $seq(limit(woo(1, n), n = k), k = 0..10)$; *# odd cosMatrix cols*

$$1, 1, 3, 15, 105, 945, 10395, 135135, 2027025, 34459425, 654729075 \quad (43)$$

> $seq(limit(woo(3, n), n = k), k = 0..10)$; *# odd cosMatrix cols*

$$\frac{2}{3}, -\frac{1}{3}, 0, 1, 10, 105, 1260, 17325, 270270, 4729725, 91891800 \quad (44)$$

> $seq(limit(woo(5, n), n = k), k = 0..10)$; *# odd cosMatrix cols*

(45)

$$\frac{2}{15}, -\frac{1}{5}, \frac{1}{15}, 0, 0, 1, 21, 378, 6930, 135135, 2837835 \quad \textbf{(45)}$$

>

$$\frac{2}{15}, -\frac{1}{5}, \frac{1}{15}, 0, 0, 1, 21, 378, 6930, 135135, 2837835 \quad \textbf{(46)}$$

>

> *# See Juggling Zeros in the Matrix (Example II)*
> *# Tom Copeland, Cheviot Hills, Los Angeles, CA, July 15 − 17, 20270*

>